

Lesson 01 - Advanced SQL

- A window function in SQL performs calculations across a set of table rows that are related to the current row

LAG - retrieves data from previous row

LEAD - retrieves data from the next row

the LAG and LEAD function shows you the data from the previous and current row at the same time / in the same row

PARTITION BY

- Partition by is used in window functions to divide your : sum(), LAG(), LEAD() set of results into groups so that the window function can operate in each group

Window - function (expression)

↳ OVER (PARTITION BY expression

ORDER BY expression)

Sales

Sale_ID	Salesperson	Sale_Date	Revenue
1	Alice	2024-01-01	1000
2	Alice	2024-01-05	1200
3	Alice	2024-01-10	900
4	Bob	2024-01-02	800
5	Bob	2025-01-09	950

SELECT Salesperson
Sale_date
revenue,

LEAD (Revenue)
OVER (PARTITION BY Salesperson
ORDER BY sale_date)
AS next_revenue
FROM sales

Sales

Sale_ID	Salesperson	Sale_Date	Revenue	next_revenue
1	Alice	2024-01-01	1000	1200
2	Alice	2024-01-05	1200	900
3	Alice	2024-01-10	900	NULL
4	Bob	2024-01-02	800	950
5	Bob	2025-01-09	950	NULL

Window Function creates a new column

```
SELECT Salesperson
       sale_date,
       revenue,
       LAG (Revenue)
         OVER (PARTITION BY Salesperson
              ORDER BY sale_date)
         AS previous_revenue
FROM sales
```

Sales

Sale_ID	Salesperson	Sale_Date	Revenue	prev_revenue
1	Alice	2024-01-01	1000	NULL
2	Alice	2024-01-05	1200	1000
3	Alice	2024-01-10	900	1200
4	Bob	2024-01-02	800	NULL
5	Bob	2025-01-09	950	800

LAG Function :

~	Sale_date	Revenue	Prev
Alice	2024-01-01	2200	NULL
Alice	2024-01-02	1700	2200

Partitions

- Partition in SQL are used with Window Functions to divide data into groups without collapsing the rows

↳ comparing values within groups without removing the rows

- 1st partition is to breakdown some into smaller groups

- A Partition by is always used with the OVER statement

NULL Functions

COALESCE (expression 1, expression 2)
 is NULL (is NOT NULL)

Replace NULL values with alternative values